

HYPOTHESIS TESTING SUMMARY

ApexPlanet Data Analytics Internship — Task 4

1. Business Question

Is there a statistically significant difference in average Total_Sales between male and female customers?

Objective: Validate whether the observed difference in average sales between the two customer groups is statistically meaningful.

2. Hypotheses

H_0 — Null Hypothesis: There is no statistically significant difference in average Total_Sales between male and female customers.

H_1 — Alternative Hypothesis: There is a statistically significant difference in average Total_Sales between male and female customers.

Significance Level (α): 0.05

3. Statistical Method

Test Used: Welch's Independent Two-Sample t-test

Test Type: Two-tailed

The test compares the average Total_Sales of two independent customer groups: Male vs Female.

Tools Used: Python, Pandas and SciPy

4. Sample Information

Metric	Male	Female
Sample Size	511	489
Mean Total Sales	141,807.34	136,883.21

Difference in Mean Sales: 4,924.13

5. Statistical Results

t-statistic: 0.6826

p-value: 0.495011

Significance Level: 0.05

Decision Rule: If p-value < 0.05, reject H_0 . If p-value $\geq$ 0.05, fail to reject H_0 .

6. Statistical Decision

p-value = 0.495011

0.495011 > 0.05 $\rightarrow$ FAIL TO REJECT H_0

There is insufficient statistical evidence to conclude that average Total_Sales differs significantly between male and female customers.

7. Conclusion

Although male customers have a higher observed average Total_Sales (141,807.34) than female customers (136,883.21), the difference is not statistically significant at the 5% significance level.

Therefore, the null hypothesis is not rejected.

8. Business Interpretation

Gender alone should not be treated as a strong basis for making major sales or marketing decisions because the observed difference was not statistically significant.

Businesses should consider additional factors such as product category, city, customer characteristics and sales trends when developing customer strategies.

9. Business Implications

- Consider product category performance.
- Consider city-level sales performance.
- Analyze customer characteristics together rather than relying on gender alone.
- Use monthly sales trends for planning.
- Validate important business assumptions statistically.

10. Final Takeaway

The observed difference in average sales between male and female customers is not statistically significant.

Data reveals the pattern. Statistical testing validates the decision.

11. Tools Used

Python • Pandas • SciPy • Microsoft Excel • Enhanced Sales Dataset